

Dynamic Demand I: Overview

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Grad IO

- ▶ Earlier this term we looked at *static models of product differentiation* such as BLP (1995).
- ▶ We have also looked at single agent models of dynamic behavior such as Rust (1987).
- ▶ What if we could put those two together? Why?

What about the following questions?

- ▶ Secondary Markets: Good or bad for sellers? overall welfare?
 - I may pay **more** for a new car today because I can sell it tomorrow.
 - I may pay **less** for a new car today because of competition with used cars.
- ▶ Durability: Should products be built to last longer or shorter?
 - I pay **more** for an appliance (dishwasher, car, microwave, iPhone, Computer, etc.) if it **lasts longer**.
 - But, I will re-purchase less frequently (Maytag Repairman)

What about the following questions?

► Adoption or Scrappage Subsidies

- Cash-For-Clunkers paid a rebate to replace your old car (Explorer/Caravan) with a new fuel-efficient one (Corolla)
- Was this about being green or about bailing out auto industry?

► Temporary Sales: Why do some products have them?

- Pure price discrimination (attracting low-value buyers?)
- Attracting switchers/ state dependence?
- Intertemporal price discrimination with storage?

Thus far we have implicitly assumed you buy a product and you receive all utility from consumption immediately. We could think about each period receiving a **flow payment** f_{ijt} . However, most of the time we could just write the NPV of future discounted flow payoffs as a **lump sum**:

$$v_{ijt} = \sum_{t=0}^{\infty} \beta^t \cdot f_{ijt}$$

and compare lump sum / NPV payoffs: v_{ijt} vs p_{jt} for different goods.

- ▶ For things like yogurt –we probably don't need to model when you choose to consume the yogurt separately from purchase.
- ▶ Maybe the good depreciates over time $f_{ijt} \geq f_{ij,t+1}$ (fine).
- ▶ Maybe it breaks with some probability ρ_t (in which case I could use “expected NPV”).
- ▶ But...

Another way to think about dynamics is to think about the rental rate:

- ▶ A house or car or other durable has a per-period price

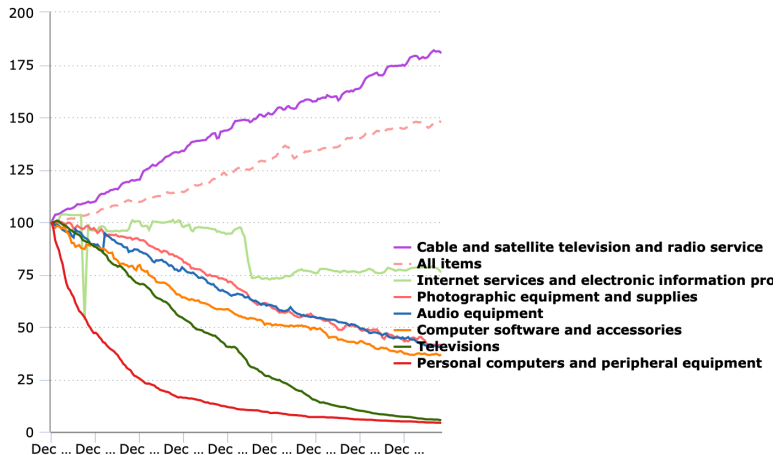
$$\Delta p_{j,t} = p_{j,t+1} - p_{j,t}$$

- ▶ You buy it and pay $p_{j,t}$ and sell it to the market at $p_{j,t+1}$ each period.
- ▶ Each period you “rent” the product to yourself at $\Delta p_{j,t}$.
- ▶ This only makes sense if the secondary market is frictionless (or we have to include a “switching” term)
 - Gavazza, Lizzeri, Roketskiy (AER 2014) do this for cars.
 - Kalouptside (AER 2014) does this to pin down the value function.

CPI: High Tech Durables

Consumer price indexes for televisions, computers, software, and related items, not seasonally adjusted, December 1997–August 2015

December 1997 = 100



- ▶ Today a 55" 4K LCD TV is \$239. In 2006, you could buy a 32" 720P TV for $> \$10,000$.
- ▶ In December 2011 TV prices fell 17% on an annual basis and other A/V equipment fell 11%, and computer equipment fell 14%.
- ▶ From August 2005 to August 2015 prices declined by 87.2%.
- ▶ We might also find that over time consumers buy better cameras or larger TV's
- ▶ The BLS tries to do *chaining* and *quality adjustments* but in high-tech products this can be very difficult.
- ▶ This has a potentially large impact on price indices (a small bias in the CPI can be billions of dollars in SSA/Medicare payments).

Adoption Curve

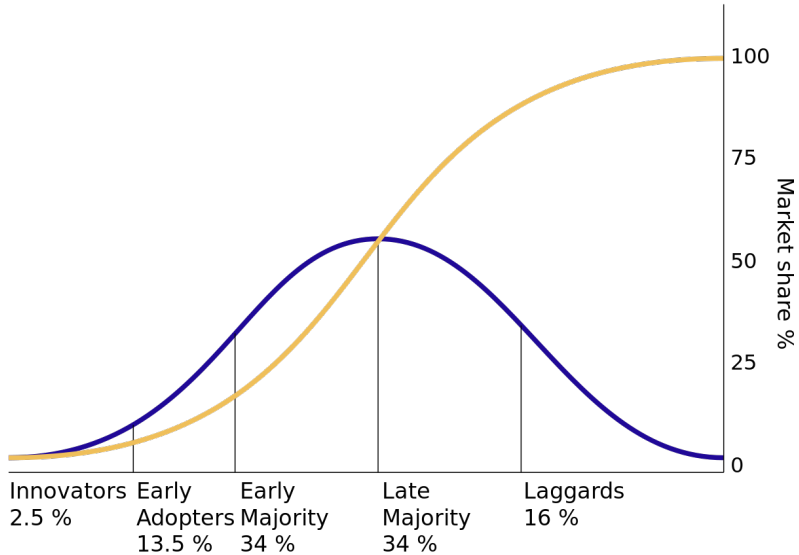


Figure 1: Average non-indicator characteristics over time

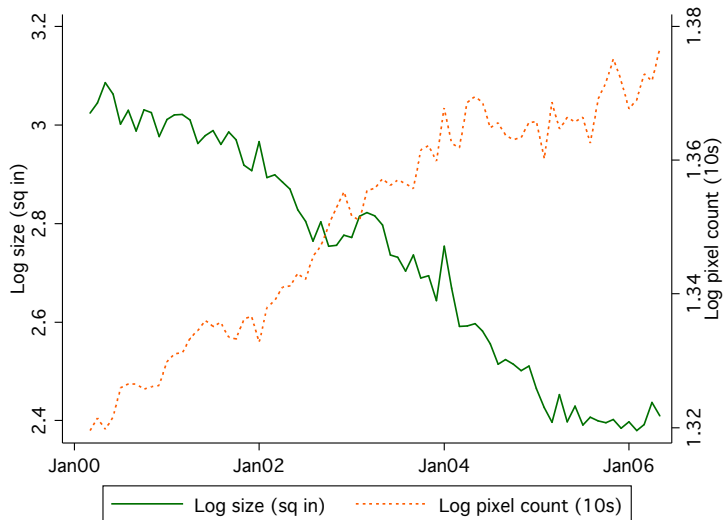
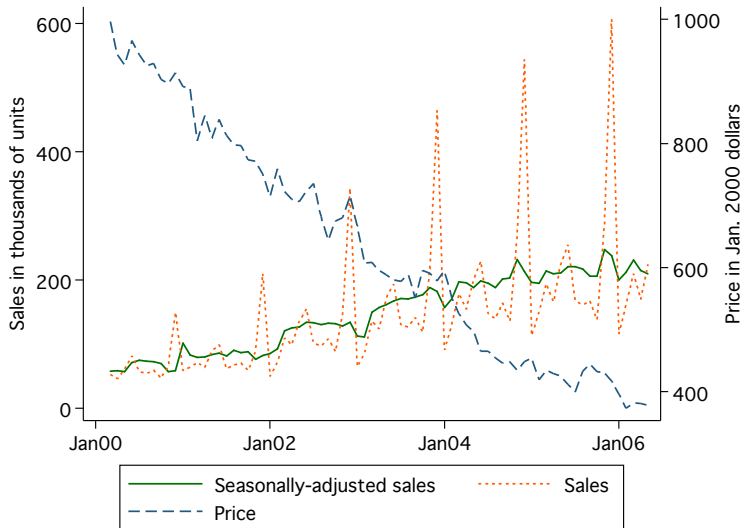


Figure 3: Prices and sales for camcorders



Imagine if we regressed P on Q (with the usual static IV):

- ▶ In early periods P falls and Q rises.
- ▶ In later periods P falls and Q also falls (top of the S -curve).
- ▶ Depending on the time period we might find that demand slopes **upwards** (lower prices lead to lower sales)

Dynamic Demand: Infeasible Static Approach

Think about this model:

$$u_{ijt} = \alpha_i x_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

$$u_{i0t} = \overline{u}_{i0t} + \varepsilon_{i0t}$$

- ▶ The real problem is that $\overline{u}_{i0t} = 0$ for all (i, t) is a bad assumption.
- ▶ If we knew $\overline{u}_{i0t}$, we could plug it in, estimate a static model and be fine.
- ▶ $\overline{u}_{i0t}$ includes several things:
 - How good is my current TV/Camera/Car today? f_{i0t} . (Initial conditions may differ)
 - What will happen tomorrow / what do I anticipate (new iPhone debut, price cuts for Black Friday, etc.)
- ▶ Idea: use the realization of $t + 1$ to inform outside option today (Rust!)

- ▶ Just proxy with a time trend or sieve (Lou Prentice Ying 2012), (Eizenberg 2011) etc. That is
$$u_{i0t} = \gamma_{0i} + \gamma_{1i}t + \gamma_{2i}t^2 + \dots$$
- ▶ We can get the elasticity correct.
- ▶ Not structural! Not helpful if we want to do counterfactuals! Can't get the elasticities under different conditions.
- ▶ Is u_{i0t} about current durable value? or Equilibrium beliefs about the future? (both!)
- ▶ Do we have i specific coefficients (we should!)

Let's start with some very strong assumptions to get our intuition clear:

1. There are $t = 1, 2$ periods.
2. Consumers can purchase at most one unit of an (infinitely) durable good
3. After purchasing the durable good they leave the market forever.

$$u_{ijt} = \alpha_i x_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

$$u_{i0t} = \varepsilon_{i0t}$$

- ▶ Suppose we estimate demand treating each period $t = 1, 2$ as a **separate market**.
- ▶ ie: close our eyes and do BLP.
- ▶ What do we get wrong?
 - How does $\frac{\partial q_{j,t}}{\partial p_{k,s}}$ look? How should it look?
- ▶ Three problems:
 - period $t = 1$ and $t = 2$ are **substitutes**
 - distribution of $f(\alpha_{it})$ is likely different in $t = 1, 2$.
 - Is $E[u_{i0t}]$ the same for all (i, t) ?

Dynamic Demand: Complete Information

Suppose consumers have full information about all shocks $(x_{j1}, x_{j2}, \xi_{j1}, \xi_{j2}, \varepsilon_{ij1}, \varepsilon_{ij2})$ in both periods.

$$u_{ijt} = \alpha_i x_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

What would we do?

- ▶ Why not estimate static demand among $\bigcup_{t=1,2} \mathcal{J}_t$ alternatives?
- ▶ Remaining issues:
 - Buying in period 1 allows me an additional period of consumption
 - Need to discount period 2 utility $\beta \times (\alpha_i x_{jt} + \xi_{jt} + \varepsilon_{ijt})$ or $\beta \times (\alpha_i x_{jt} + \xi_{jt}) + \varepsilon_{ijt}$
 - A single outside good of not purchasing in either period $u_{i0} = \varepsilon_{i0}$
- ▶ How bad is this model? Compared to the last one?

Dynamic Demand: Relaxing some assumptions

Problematic assumption was probably full information. Suppose instead that only $(\varepsilon_{ij2}, \varepsilon_{i0})$ are **unobserved** in $t = 1$.

$$v_{i0,t=1} \equiv \mathbb{E}_{\varepsilon}[\max_j u_{ij2} | \Omega_{t=1}] = \log \left(\sum_j \exp[\alpha_i x_{j2} + \xi_{j2}] \right) + \eta$$

- ▶ $\Omega_{t=1} = \{x_{j1}, x_{j2}, \xi_{j1}, \xi_{j2}, \varepsilon_{ij1}\}$ (everything known at $t = 1$)
- ▶ η is Euler's constant (per usual).
- ▶ What about outside good?
 - Either just another good in $t = 2$ with $\alpha_i x_{j2} + \xi_{j2} = 0$
 - Or we consider $v_{i0,t=2} = \mathbb{E}[\max\{\varepsilon_{i0}, \max_j u_{ij2}\} | \Omega_{t=1}]$

Dynamic Demand: What about Beliefs?

- ▶ This assumed that x_{j2} and ξ_{j2} were observed in $\Omega_{t=1}$.
- ▶ Maybe we want to make some component of x_{j2} **unobserved** (such as price or ξ).

$$\mathbb{E}_t \left[\log \left(\sum_j \exp[\alpha_i p_{j2} + \xi_{j2}] \right) | \Omega_t \right] = \int \log \left(\sum_j \exp[\alpha_i p_{j2} + \xi_{j2}] \right) g(\mathbf{p}_2 | \Omega_t)$$

Integrate out over the unknown (distribution depends on information Ω_t)

$$\mathbb{E}_t \left[\log \left(\sum_j \exp[\alpha_i x_{j2} + \xi_{j2}] \right) \mid \Omega_t \right] = \log \left(\sum_j \exp[\alpha_i x_{j2} + \xi_{j2}] \right) + \eta + \zeta_{it}$$

Rational expectations implies that expectational error ζ_{it} is orthogonal to everything known at Ω_t

► Can't use anything in Ω_t to predict ζ_{it} so $\mathbb{E}_t[\zeta_{it} \times A(\Omega_t)] = 0$

Dynamic Demand: Relaxing some assumptions

Now what?

$$u_{ij,t=1} = \alpha_i x_{j1} + \xi_{j1} + \varepsilon_{ij1}$$

$$u_{i0,t=1} = \bar{u}_{i0,t=1} + \beta v_{i0,t=1} + \zeta_{i,t=1} + \varepsilon_{i01}$$

$$u_{ij,t=2} = \alpha_i x_{j2} + \xi_{j2} + \varepsilon_{ij2}$$

$$u_{i0,t=2} = \bar{u}_{i0,t=2} + \underbrace{\beta v_{i0,t=2} + \zeta_{i,t=2}}_{=0} + \varepsilon_{i02}$$

- ▶ If $(\bar{u}_{i0,t}, \beta v_{i0,t=1})$ are known we can estimate static demand with each t as a separate “market”.
- ▶ I pulled an extra ε_{i01} out of my hat.
- ▶ What is the other dynamic linkage?

Dynamic Demand: What about cream skimming?

Also need to account for the fact that $f(\alpha_{i,t=1})$ and $f(\alpha_{i,t=2})$ are not the same:

- ▶ If goods are perfectly durable, and consumers permanently exit the market...
- ▶ Assume that $f(\alpha_{i,t=1}) = w_{i,t=1}$ is a discrete distribution of “types”
 - Note: “type” does not include ε .
 - then $w_{i,t=2} = w_{i,t=1} \cdot s_{i0,t=1}$

Dynamic Demand: Can we go further?

Replacement / Upgrades:

- ▶ Suppose we allow the people who purchase in $t = 1$ to remain in the market.
- ▶ Now part of your “type” α_i includes your existing stock of the durable good $\bar{u}_{i0t}$
 - Think about as a RC on the constant β_{i0}
- ▶ A purchase increases the value of outside good (previously to ∞)
- ▶ The transitions become more complicated $w_{t=2} = f(w_{t=1}, s_{ij,t=1})$.
 - You still throw the old durable into the trash when you are done.
 - Could also allow for a scrap value.
 - Need to be a bit careful about NPV of expected stream of payments vs. “flow utility” now.

- ▶ Same idea:

$$u_{ijt} = \alpha_i x_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

$$u_{i0t} = \bar{u}_{i0t} + \mathbb{E}[\max_j u_{ij,t+1} | \Omega_t] + \varepsilon_{i0t}$$

- ▶ My type $\bar{u}_{i0t}$: how much laundry detergent I have left.
- ▶ My beliefs $\mathbb{E}[\max_j u_{ij,t+1} | \Omega_t]$: is my preferred brand likely to be on discount in the near future?
- ▶ Bias from $u_{i0t} = \varepsilon_{i0t}$. Sales are high during discounts and low following discounts. We think this implies demand is **too elastic** relative to a permanent price change.
- ▶ Durables $Corr(u_{i0t}, p_{jt}) < 0$ (time trend).
Storables: $Corr(u_{i0t}, p_{jt}) > 0$ (sale in adjacent period).

Thanks!
